Sidhant Singhvi

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Summary

Computer Engineering student at UC San Diego with 4+ years of experience in hardware and embedded systems with a strong curiosity for hands-on, system-level engineering. Experienced in building ML-driven systems across distributed compute, and data-intensive pipelines. Winner of HardHack.

Education

University of California San Diego
Bachelor of Science in Computer Engineering — September 2022 - June 2026

Experience

Systems Engineering Intern — July 2025 – Present
BRG Energies — San Diego

- Integrated Taskyon into BRG's energy platform, enabling automated workflows and improving compute resource efficiency by 28% for renewable energy analysis.
- Enabled in-browser execution of renewable energy simulations for evaluation of solar/wind generation runs.
- Implemented client-side state management to support parcel-level forecasting and visualization.

Undergraduate Researcher – Embedded Firmware and Systems — September 2025 – Present
Engineers for Exploration, UCSD — San Diego

- Developing an embedded control system for a low-power radio telemetry drone used in wildlife monitoring.
- Designing firmware for communication over I2C, UART, and GPIO to interface with sensors and RF modules.
- Contributed to PCB design for field-deployable devices with efficient solar and battery power management.

Embedded Firmware Engineering Intern — May 2025 – July 2025
Qualcomm Institute — San Diego

- Formulated real-time firmware pipelines for solar weather compute systems, enforcing deterministic execution windows to sustain VR visualization workloads at 60+ FPS under latency constraints.
- Engineered UART and SPI sensor interfaces and drivers, achieving 99% runtime reliability.
- Applied multithreading, optimizing HPC embedded systems with 25% faster compute on CPU baselines.

Projects

Autonomous Navigation System with Embedded Control — April 2025

- Developed an autonomous vehicle navigation system integrating GNSS, IMU, and camera sensors using ROS2.
- Designed and refined an extended Kalman filter, reducing position error by 22% in real-world navigation tests.
- Built a custom control board enabling real-time communication over I2C, UART, and CAN.

EyeSpy| *Winner, Hardware Hackathon* — January 2025

- Spearheaded an object recognition–based dementia care app, coordinating hardware–software prototyping.
- Enhanced real-time object recognition with hardware acceleration, achieving 1.4× faster inference speeds.
- Engineered state retention logic, enabling context handoff for dementia-assisted interaction.

CUDA N-Body Simulation – GPU Computing — October 2024

- Simulated gravitational interactions between bodies using both CPU and CUDA-based GPU parallelization.
- Achieved 60x speedup in execution time with CUDA parallelization, reducing CPU time from 0.91s to 0.015s.
- Optimized force calculations with thread-per-body execution on the GPU, enhancing performance.

Hardware-Accelerated Error Detection Circuit — September 2024

- Designed a multi-phase error detection circuit using Verilog to improve data integrity in communication.
- Implemented SEC-DED Hamming code, achieving 100% single-bit correction and double-bit detection.
- Optimized circuit complexity using minimization techniques such as Karnaugh maps and Quine-McCluskey.

Technical Skills

Languages: C++, Python, Java, C, CUDA, Verilog, OpenCL, Arduino, Assembly, JavaScript/TypeScript
Embedded Systems and Robotics: Jetson Nano, GNSS-based navigation, Embedded Linux, Remote Access
Circuit Design Tools: SPICE, LTspice, CAD, Digital, Cadence Virtuoso
Frameworks/Libraries: Node.js, OpenCV, TensorFlow, PyTorch
Other Skills: Leadership, Web/Android/iOS, Data Analysis, Microsoft Office, MATLAB
Lab Tools: Soldering, Prototyping, Oscilloscope, Impedance Analyzer, Power Supplies, PCB Assembly, 3D Print
University Courses: Computer Architecture, Algorithms and Data Structures, Advanced Digital Design, Linear Systems, Operating Systems, Linear and Nonlinear Optimization, Engineering Probability, Deep Learning, AI with Probabilistic Reasoning, Object-Oriented Design, Analog Design, Product Design and Entrepreneurship